DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 62

Question description

There are three common ways to traverse the nodes of a binary tree:

- *Preorder*: First process the root, then the left subtree, and finally the right subtree.
- *Inorder*: First process the left subtree, then the root, and finally the right subtree.
- *Postorder*: First process the left subtree, then the right subtree, and finally the root.

There is a binary tree of n nodes with distinct labels. You are given the preorder and inorder traversals of the tree, and your task is to determine its postorder traversal.

Constraints

- $1 \leq n \leq 10^5$

Input

The first input line has an integer n : the number of nodes. The nodes are numbered 1,2,…,n.

After this, there are two lines describing the preorder and inorder traversals of the tree. Both lines consist of n integers.

You can assume that the input corresponds to a binary tree.

Output

Print the postorder traversal of the tree.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5
5 3 2 1 4
3 5 1 2 4
```

EXPECTED OUTPUT

```
3 1 4 2 5
```

Test Case 2

INPUT (STDIN)

```
6
6 5 3 2 1 4
6 3 5 1 2 4
```

EXPECTED OUTPUT

```
3 1 4 2 5 6
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(i=1;i<=n;i++)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
8
```

Test Case 2

TOKEN COUNT

```
390
```

Test Case 3

NLOC

```
41
```

Code

solution.c

```c
#include <stdio.h>

int search(int arr[], int x, int n) {
    for (int i = 0; i < n; i++) if (arr[i] == x) return i;
    return -1;
}

void printPostOrder(int in[], int pre[], int n) {
    int root = search(in, pre[0], n);
    if (root != 0) printPostOrder(in, pre + 1, root);
    if (root != n - 1) printPostOrder(in + root + 1, pre + root + 1, n - root - 1);
    printf("%d ", pre[0]);
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int pre[n], in[n];
    for (int i = 0; i < n; i++) scanf("%d", &pre[i]);
    for (int i = 0; i < n; i++) scanf("%d", &in[i]);
    printPostOrder(in, pre, n);
    printf("\n");
    return 0;
}
```

Output

Problems Output Debug Console Terminal Ports powershell

```
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
5
5 3 2 1 4
3 5 1 2 4
3 1 4 2 5
PS C:\C Programming>
```